

Module 5- Week 8

IO Programming-Part2

Winter 2023

Topics

- Interfacing to an LCD
- Interfacing to a Keyboard

Pin Descriptions for LCD

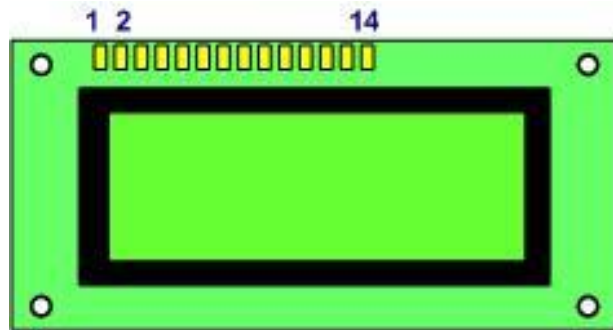
Power

Control

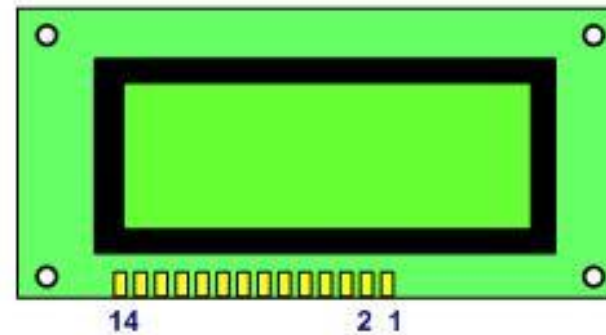
Data

Pin	Symbol	I/O	Description
1	VSS	--	Ground
2	VCC	--	+5V power supply
3	VEE	--	Power supply to control contrast
4	RS	I	RS = 0 to select command register, RS = 1 to select data register
5	R/W	I	R/W = 0 for write, R/W = 1 for read
6	E	I	Enable
7	DB0	I/O	The 8-bit data bus
8	DB1	I/O	The 8-bit data bus
9	DB2	I/O	The 8-bit data bus
10	DB3	I/O	The 8-bit data bus
11	DB4	I/O	The 4/8-bit data bus
12	DB5	I/O	The 4/8-bit data bus
13	DB6	I/O	The 4/8-bit data bus
14	DB7	I/O	The 4/8-bit data bus

Pin Positions for Various LCDs from Optrex



DMC1610A
DMC1606C
DMC16117
DMC16128
DMC16129
DMC1616433
DMC20434

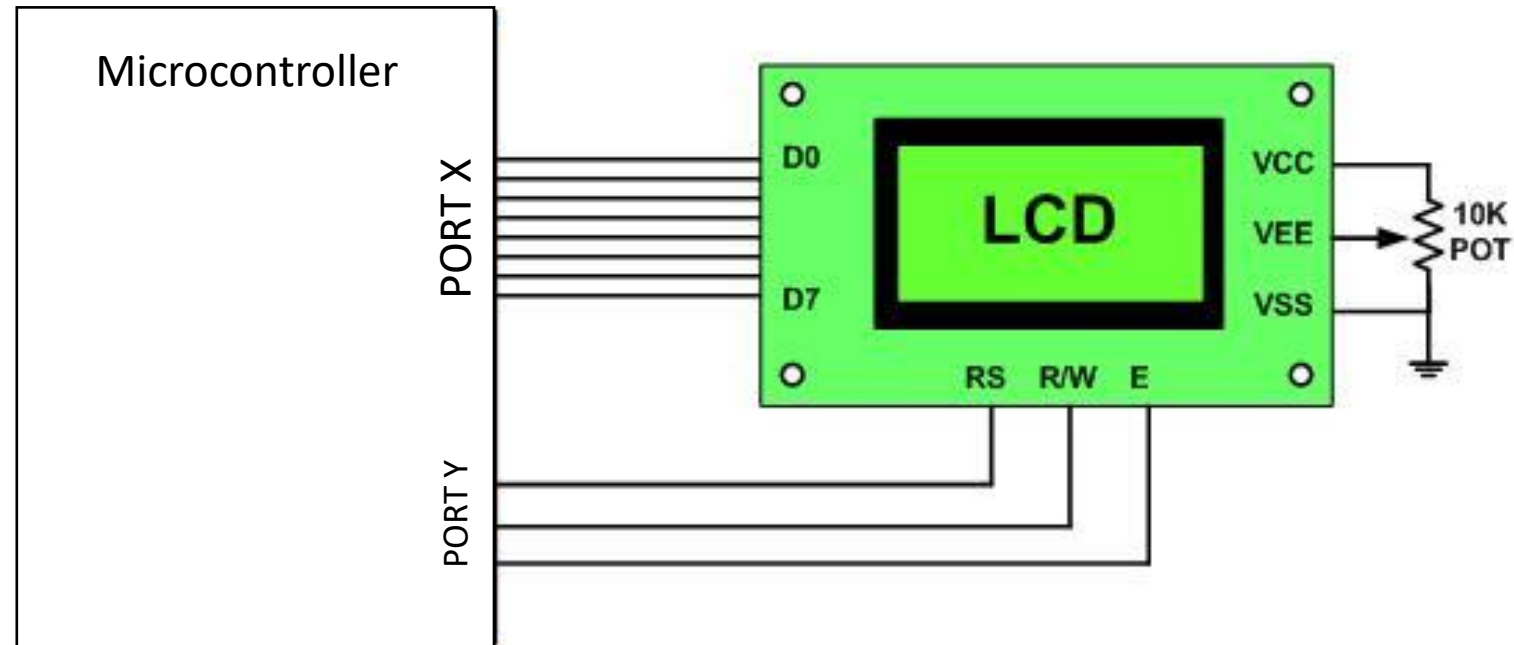


DMC16106B
DMC16207
DMC16230
DMC20215
DMC3216



DMC20261
DMC24227
DMC24138
DMC32132
DMC32239
DMC40131
DMC40218

LCD Connection to Microcontroller



RS, Register Select

- There are two registers inside the LCD and the RS pin is used for their selection.
 - If $RS = 0$, the instruction code register is selected, allowing the user to send a command.
 - If $RS = 1$, the data register is selected, allowing the user to send data to be displayed on the LCD (or to receive data from the LCD controller).

E, Enable

- The enable pin is used by the LCD to latch information presented to its data pins .
- When data is supplied to data pins, a HIGH-to-LOW pulse must be applied to this pin for the LCD to latch in the data present at the data pins.
- The pulse must be a minimum of 230ns wide according to the Hitachi datasheet.

R/W, read/write

- R/W input allows the user to write information into the LCD controller or read information from it.
 - $R/W = 1$, when reading
 - $R/W = 0$, when writing

D0 –D7

- The 8-bit data pins are used to send information to the LCD or read the content of the LCD's internal registers.
- The LCD controller can also operate with 4-bit data and only D4-D7 are used.
- To display letters, numbers, and characters, we send ASCII codes to these pins while RS = 1.

Some commonly used LCD Command Codes

Code (Hex)	Command to LCD Instruction Register
1	Clear display screen
2	Return cursor home
6	Increment cursor (shift cursor to right)
F	Display on, cursor blinking
80	Force cursor to beginning of 1st line
C0	Force cursor to beginning of 2nd line
38	2 lines and 5x7 character (8-bit data, D0 to D7)
28	2 lines and 5x7 character (4-bit data, D4 to D7)

List of LCD Instructions

Execution Time for
Each Command

Instruction	RS	R/ W	DB 7	DB 6	DB 5	DB 4	DB 3	DB 2	DB 1	DB 0	Description	Execution Time (Max)
Clear display	0	0	0	0	0	0	0	0	0	1	Clears entire display and sets DD RAM address 0 in address counter	1.64 ms
Return Home	0	0	0	0	0	0	0	0	1	-	Sets DD RAM address to 0 as address counter. Also returns display being shifted to original positions. DD RAM contents remain unchanged.	1.64 ms
Entry Mode Set	0	0	0	0	0	0	0	1	I/D	S	Sets cursor move direction and specifies shift of display. These operations are performed during data write and read.	40μs
Display On/Off Control	0	0	0	0	0	0	1	D	C	B	Sets On/Off of entire display (D), cursor On/Off (C), and blink of cursor position character (B).	40μs
Cursor or Display shift	0	0	0	0	0	1	S/C	R/L	-	-	Moves cursor and shifts display without changing DD RAM contents.	40μs
Function Set	0	0	0	0	1	DL	N	F	-	-	Sets interface data length (DL), number of display lines (L), and character font (F)	40μs
Set CG RAM Address	0	0	0	1	AGC						Sets CG RAM address. CG RAM data is sent and received after this setting.	40μs
Set DD RAM Address	0	0	1	ADD						Sets DD RAM address. DD RAM data is sent and received after this setting.	40μs	
Read Busy Flag & Address	0	1	BF	AC						Reads Busy flag (BF) indicating internal operation is being performed and reads address counter contents.	40μs	
Write Data CG or DD RAM	1	0	Write Data						Writes data into DD or CG RAM.			40μs
Read Data CG or DD RAM	1	1	Read Data						Reads data from DD or CG RAM.			40μs

List of LCD Instructions (Abbreviations)

- DD RAM: Display data RAM
- CG RAM: Character generator RAM
- AGC: CG RAM address
- ADD: DD RAM address, corresponds to cursor address
- AC: address counter used for both DD and CG RAM addresses
- I/D: 1 = Increment, 0: Decrement
- S =1: Accompanies display shift
- S/C: 1 = Display shift, 0: Cursor move
- R/L: 1: Shift to the right, 0: Shift to the left
- DL: 1 = 8 bits, 0 = 4 bits
- N: 1 = 2-line, 0 = 1-line
- F: 1 = 5 x 10 dots, 0 = 5 x 7 dots
- BF: 1 = Internal operation, 0 = Can accept instruction

Steps to Configure LCD

- **Step 1: Initialize GPIO for LCD**

- Enable **Port A** and **Port B**.
- Configure **Port B** and **PA7, PA6, PA5** as outputs.

- **Step 2: Initialize the LCD**

1. Wait for **Power-On Delay** (about 20ms).
2. Send the **Function Set** command to enable 4-bit mode.
3. Set **Display Control** (Turn ON display, cursor, blinking).
4. Clear Display and Set Entry Mode.

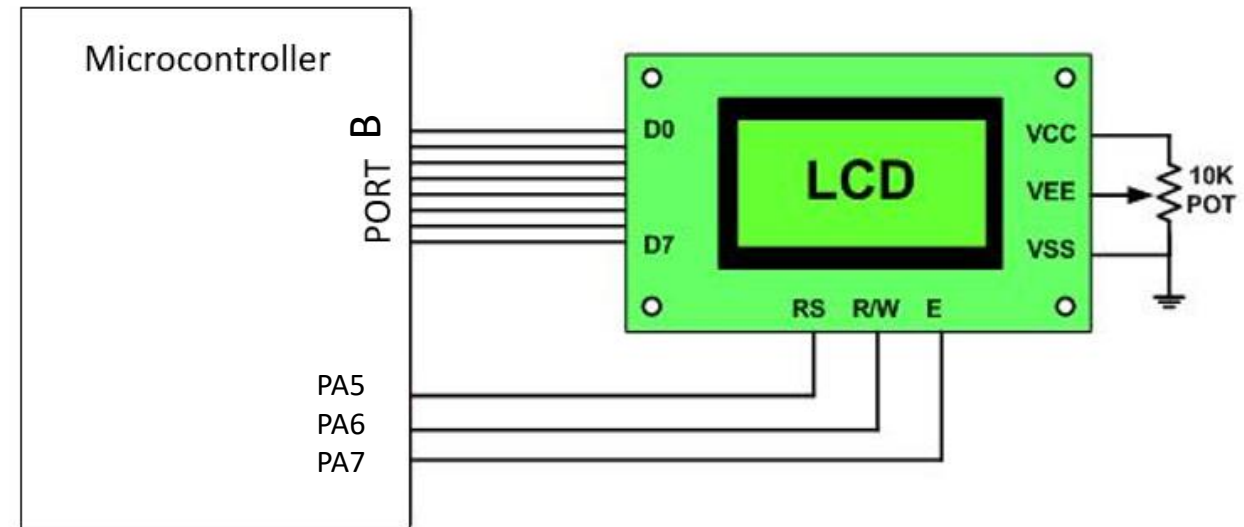
- **Step 3: Send Commands and Data**

- Commands (like clearing display, setting cursor position) are sent with **RS = 0**.
- Data (characters to display) are sent with **RS = 1**.
- Send data in **two nibbles** (higher 4 bits first, then lower 4 bits).
- Pulse **Enable (E) Pin** to latch data.

Configuring Pins and Controls for LCD

```
#include "TM4C123GH6PM.h"

#define LCD_DATA GPIOB
#define LCD_CTRL GPIOA
#define RS 0x20 /* PORTA BIT5 mask */
#define RW 0x40 /* PORTA BIT6 mask */
#define EN 0x80 /* PORTA BIT7 mask */
```



Sending commands to LCDs

- To send any commands to the LCD:
 - make pins RS = 0 and R/W = 0 and
 - send HIGH-to-LOW pulse on the E pin to enable the internal latch of the LCD.
 - wait for the length of execution time required for the command by incorporating delay in your code.

```

void LCD_command(unsigned char command)
{
    LCD_CTRL->DATA = 0;      /* RS = 0, R/W = 0 */
    LCD_DATA->DATA = command;
    LCD_CTRL->DATA = EN;     /* pulse E */
    delayUs(0);
    LCD_CTRL->DATA = 0;
    if (command < 4)
        delayMs(2);          /* command 1 and 2 needs up to 1.64ms */
    else
        delayUs(40);         /* all others 40 us */
}

```

Sending Data to the LCD

- In order to send data to the LCD to be displayed, we must:
 - set pin RS = 1 and R/W = 0.
 - Put Data on LCD's DATA port.
 - Send HIGH-to-LOW pulse on the E pin to enable the internal latch of the LCD.

```
void LCD_data(unsigned char data)
{
    LCD_CTRL->DATA = RS;    /* RS = 1, R/W = 0 */
    LCD_DATA->DATA = data;
    LCD_CTRL->DATA = EN | RS; /* pulse E */
    delayUs(0);
    LCD_CTRL->DATA = 0;
    delayUs(40);
}
```

There are five main functions in the code:

```
void delayMs (int n);  
void delayUs (int n);  
void LCD_command(unsigned char command);  
void LCD_data(unsigned char data);  
void LCD_init(void);
```

LCD Initialization

- Refer to the link below for more details:

[LCD Initialization](#)

```

void LCD_init(void)
{
    SYSCTL->RCGCGPIO |= 0x01; /* enable clock to GPIOA */
    SYSCTL->RCGCGPIO |= 0x02; /* enable clock to GPIOB */

    LCD_CTRL->DIR |= 0xE0; /* set PORTA pin 7-5 as output for control */
    LCD_CTRL->DEN |= 0xE0; /* set PORTA pin 7-5 as digital pins */
    LCD_DATA->DIR = 0xFF; /* set all PORTB pins as output for data */
    LCD_DATA->DEN = 0xFF; /* set all PORTB pins as digital pins */

    delayMs(20); /* initialization sequence */
    LCD_command(0x30);
    delayMs(5);
    LCD_command(0x30);
    delayUs(100);
    LCD_command(0x30);

    LCD_command(0x38); /* set 8-bit data, 2-line, 5x7 font */
    LCD_command(0x06); /* move cursor right */
    LCD_command(0x01); /* clear screen, move cursor to home */
    LCD_command(0x0F); /* turn on display, cursor blinking */
}

```

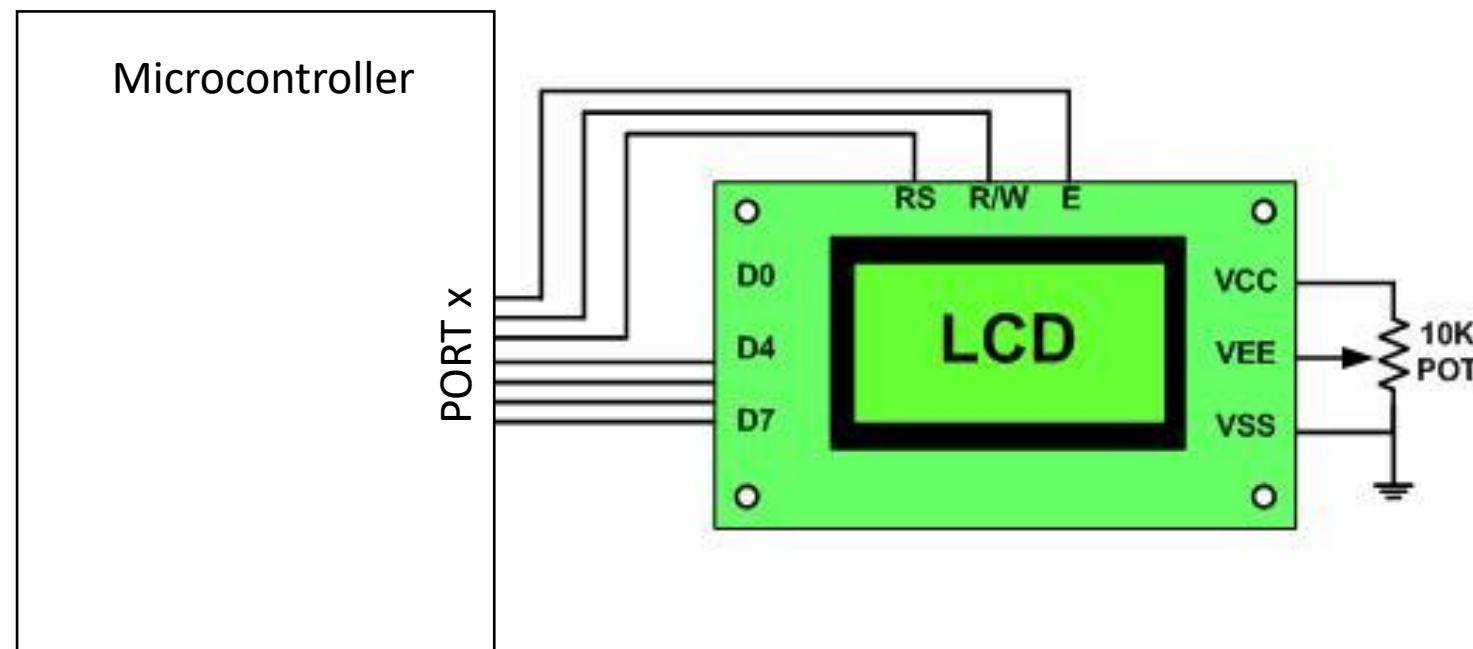
C:\Users\N01338953\OneDrive - Humber College\Documents\TIVA G-Series\LAB4\LCD.uvprojx - µVision
[Non-Commercial Use License]

Checking LCD busy flag

- In the previous code, a time delay is used before issuing the next data or command. This allows the LCD enough time to get ready to accept the next data.
- LCD, however, has a busy flag. We can monitor busy flag and issue data.
- The busy flag is the D7 bit of command register. To check the busy flag, we must read the command register (R/W = 1, RS = 0)
- When the LCD is busy, D7 = 1.
- When the LCD is ready, D7 = 0.

LCD Connection for 4-bit Data

To save number of microcontroller pins used by LCD interfacing, we can use the 4-bit data option instead of 8-bit. In this case, we only connect D7- D4. Including the three control pins, we can use only one 8-bit port to interface with LCD.



LCD Cursor Position

In the LCD, one can move the cursor to any location in the display by issuing an address command. The next character sent will appear at the cursor position.

LCD Addressing Command:

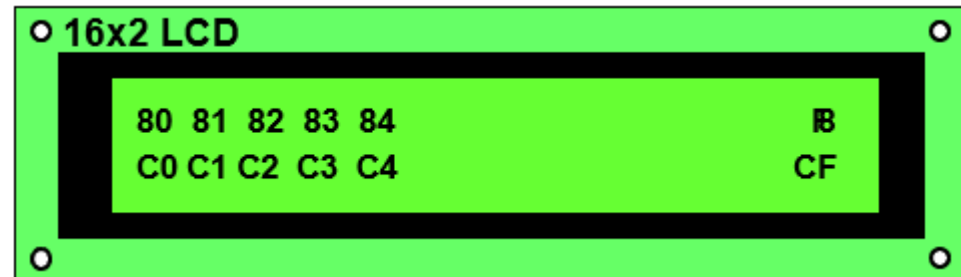
	DB7	DB6	DB5	DB4	DB3	DB2	DB1	DB0
Line 1 (min)	1	0	0	0	0	0	0	0
Line 1 (max)	1	0	1	0	0	1	1	1
Line 2 (min)	1	1	0	0	0	0	0	0
Line 2 (max)	1	1	1	0	0	1	1	1

LCD Cursor Position Examples

For the two-line LCD, the address command for the first location of line 1 is 0x80, and for line 2 it is 0xC0.

Set cursor at the fourth location of line 1: `LCD_command(0x83)`

Set cursor at the sixth location of line 2: `LCD_command(0xC5)`



Reference

1. Mazidi, M., Chen, S., Naimi, S., Naimi, S., Chen, S. (2016). TI TIVA ARM Programming For Embedded Systems, 1st Edition. MicroDigitalEd. ISBN-13: 978-0997925920